NLP COURSE PROJECT

Error Analysis of POS Taggers

A Systematic Comparison of NLTK · spaCy · Stanza · BERT on Universal Dependencies (English EWT)

4	~25K	17	7
Taggers	Test Tokens	UPOS Tags	Error Dimensions

Survey — Literature Review

Universal Dependencies

Nivre et al. (2016)

Cross-lingual treebank framework; defines the 17 UPOS tags used as gold standard throughout this project.

BERT Transformers

Devlin et al. (2019)

Pre-trained deep bidirectional transformer; fine-tuned on UD for POS; sets state-of-the-art on sequence labelling.

Stanza NLP

Qi et al. (2020)

BiLSTM+CRF pipeline trained natively on UD treebanks; strong multilingual coverage and UD-aligned output.

Contextual Embeddings

Akbik et al. (2018)

Character-level language model embeddings; contextualizes rare/OOV words better than static word vectors.

Implementation

1. **NLTK Avg. Perceptron**
   Statistical — Penn TB → UPOS

2. **spaCy en_core_web_sm**
   CNN Neural — direct UPOS

3. **Stanza BiLSTM + CRF**
   Neural — trained on UD

4. **BERT (HuggingFace)**
   Transformer — fine-tuned UD

1. Load UD English EWT (CoNLL-U)
2. Build vocabulary & OOV stats from train split
3. Wrap all 4 taggers in a uniform interface
4. Tag 25K test tokens; align predictions
5. ErrorAnalyzer per tagger + CrossTaggerAnalysis
6. Generate 15+ plots → results/

Analysis

~89%	~93%	~95%	~97%
NLTK	spaCy	Stanza	BERT

Per-Tag F1

PROPN & PUNCT hardest across all taggers; BERT leads on ambiguous categories.

Confusion Matrix

NOUN↔PROPN and VERB↔AUX are most frequent confusion pairs.

OOV Analysis

All taggers degrade on out-of-vocabulary words; BERT degrades the least.

Frequency Buckets

Error rate inversely correlates with token frequency across all taggers.

Lexical Ambiguity

Ambiguous words drive 40–60% of errors; OOV × ambiguity = highest error.

Sentence Position

Error rates slightly higher at sentence boundaries and in rare positions.

Innovation

Majority-Vote Ensemble

Combines all 4 taggers via majority vote. Ensemble accuracy exceeds best individual tagger by 1–2%, showing complementary strengths.

Hard-Case Mining

Identifies tokens where ALL taggers fail simultaneously — revealing intrinsically ambiguous or rare constructions that no current model handles well.

5-Class Error Taxonomy

Every error labelled as: OOV · Lexical Ambiguity · Proper Noun · Punctuation · Other. Enables targeted improvement strategies per tagger.

Pairwise Agreement Analysis

Cross-tagger agreement matrix shows BERT+Stanza agree most (>96%), while NLTK diverges most, quantifying model diversity.

Key Takeaways

Architecture Matters:
BERT >> Stanza >> spaCy >> NLTK. Transformer-based models outperform CNN and statistical approaches by a clear margin.

OOV is the Core Challenge:
All taggers degrade significantly on unseen words. BERT's contextual representations mitigate this best.

Ensemble Wins:
Majority-vote combining all 4 taggers outperforms any single tagger, exploiting model diversity.

PROPN & Punctuation are Hard:
Proper nouns and punctuation remain problematic even for BERT due to naming conventions and context dependency.

Taxonomy Drives Improvement:
Classifying errors into 5 types reveals actionable targets: OOV → better subword; Ambiguity → more context; PROPN → named-entity pre-training.